

V_0 : Raw Verifier Output

```
(((((... == 37 && n == n && 1369 ==  
((long long) n * n)) || (37 < n  
|| 44 == i && n == n && 1936 ==  
((long long) n * n)) || (44 < n  
|| sum == 1764 && 42 == i && 42 < n)  
|| (1764 == ((long long) n * n  
|| sum == 0 && n == n && 0 <= n  
:  
:  
|| i == 54 && n == n && 2916 ==  
((long long) n * n)) || (54 < n
```

12,384 chars, 162 disjuncts

V_1 : Normalized

```
i == 37 && sum == 1369 &&  
1369 == n * n || 37 < n  
|| 44 == i && sum == 1936  
&& 1936 == n * n || 44 < n  
|| sum == 1764 && 42 == i  
&& 42 < n || 1764 == n * n  
|| sum == 0 && i == 0 && 0  
<= n  
:  
:  
|| i == 54 && sum == 2916  
&& 2916 == n * n || 54 < n
```

6,780 chars, 162 disjuncts

normalize

LLM

simplify

V_2 : Simplified

```
0 <= i && i <= n  
&& sum == i*i
```

30 chars, 3 conjuncts